Pure-Information Determination of the Hehl–Datta Coefficient: Fisher–Canonical Matching Closed by the Null-Plane Limit and Finite-Ball Absorbability

Robert W. McGwier
CTO and Staff Scientist, Cohere Technology Group, Herndon, VA

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Abstract

Paper V reduced the fully information-theoretic fixing of the contortion stiffness c_2 (and therefore of the Hehl–Datta coefficient) to an overdetermined Fisher–canonical matching equation whose kinematic core is the universal integral $\mathcal{I}_B$. With $\mathcal{I}_B$ now established to be non-vanishing, that matching remained the last open item in the axial channel. The present note closes it. On null planes $\mathcal{I}_B \equiv 0$ by the Markov property; the matching collapses to a purely local algebraic relation that forces $c_2 = 3/\kappa$. For finite balls the non-vanishing residue is absorbed by the already-proved dictionary improvement of Paper IV, so the same value of c_2 continues to satisfy the matching for every ball. The pure-information magnitude of the Hehl–Datta interaction is therefore

$$\mathcal{L}_{\text{HD}} = -\frac{3\kappa}{16}\, \mathcal{J}_5 \cdot \mathcal{J}_5.$$

No classical variational principle is used. Every material claim is labeled **[P]**, **[C]** or **[O]**.

Why this work is required. The Einstein–Cartan certification of the New Standard Model from entanglement bookkeeping is incomplete until the absolute scale of its single new interaction is derived from quantum-information principles alone. Paper V obtained the sign from relative-entropy positivity and the magnitude by rigidity under program normalizations; the pure-information route was left open. Closing the Fisher–canonical matching removes the final obstruction and converts the Hehl–Datta coefficient into a theorem of entanglement thermodynamics.

Epistemic Conventions

Every material claim is tagged: **[P]** proved or derived within a stated framework; **[C]** conjectured with nontrivial support; **[O]** open. Numerical agreement is not proof. Consensus is not proof.

1 The Residual Matching

Paper V decomposed the Hehl–Datta coefficient into three layers of decreasing strength:

1. **Sign [P]**: relative-entropy positivity, via Fisher information = canonical energy (Lashkari–Van Raamsdonk), forces positive contortion stiffness and therefore the negative contact sign.
2. **Magnitude by rigidity [P]**: given the Ryu–Takayanagi normalization (unconditional by Paper IV) and the modular selection rule excluding independent torsion invariants, the coefficient is uniquely $-3\kappa/16$, with the pure numbers extracted from gamma-matrix and epsilon-tensor first principles.
3. **Magnitude from pure information [O]**: treat c_2 as unknown and impose the channel-by-channel equality of boundary Fisher information with bulk canonical energy,

   $$\frac{1}{2}\,\mathcal{F}_B[\beta] \;=\; \mathcal{E}^{\text{axial}}_{\text{can},B}\big[K(\beta;c_2),A(\beta)\big] \qquad \forall \text{ balls } B. \tag{1}$$

   The kinematic content of (1) is identical to the universal functional $\mathcal{I}_B$ of Paper IV. Evaluating that structure was the designated remaining computation.

Paper VI established $\mathcal{I}_B \not\equiv 0$ for finite regions and activated the absorbability proposition. The matching equation itself is the subject of the present note.

2 Why the Evaluation Is Required

The New Standard Model of the program contains exactly one interaction beyond the Standard Model and Einstein gravity: the universal axial–axial contact term obtained by eliminating torsion. Its coefficient must be fixed by the same entanglement bookkeeping that produces the Einstein equations if the claim that gravity is entanglement thermodynamics is to be complete. The rigidity argument already yields the classical value, but still invokes the Palatini form of the gravitational action as an input (through the selection rule that forbids independent torsion kinetic terms). A pure-information derivation removes that residual classical assumption and places the numerical coefficient on the same footing as the sign: a theorem of relative entropy and modular kinematics.

3 Null-Plane Limit

On regions whose future horizon lies on a null plane the vacuum is Markov [9]. Modular Hamiltonians of null cuts are strictly local, every non-zero modular-frequency matrix element vanishes, and therefore

$$\mathcal{I}_B \equiv 0.$$

The modular response of a smooth axial source β reduces to a purely local, boost-weighted term—the direct analogue of the Faulkner–Leigh–Parrikar–Wang result for stress-tensor deformations. In that limit the Fisher–canonical matching (1) collapses to an algebraic relation between the local axial source and the induced contortion. The unique constant stiffness consistent with

- the Ryu–Takayanagi coefficient $1/4G_N$ already fixed by the metric-sector first law, and
- positivity of bulk canonical energy

is the Palatini value

$$c_2 = \frac{3}{\kappa}. \tag{2}$$

(The same arithmetic that appears in the rigidity theorem is recovered, now from the null-plane matching alone.) Completing the square then yields

$$\mathcal{L}_{\mathrm{HD}} = -\frac{3\kappa}{16}\,\mathcal{J}_5 \cdot \mathcal{J}_5.[\mathbf{P}]$$

4 Finite Balls

For a finite ball $\mathcal{I}_B \neq 0$. The residue generates an additional finite, universal, state-independent bilinear form linear in the axial source. By the absorbability proposition of Paper IV this form is absorbed into a redefinition of the local modular term—equivalently, a finite improvement of the holographic dictionary that maps the boundary axial source to the bulk contortion boundary value. After that improvement the finite-ball matching is identical in structure to the null-plane matching. Consequently the same value of c_2 continues to satisfy (1) for every ball.

Because the family of equations (one per ball) is overdetermined, and because the null-plane members already force $c_2 = 3/\kappa$, no other constant stiffness is consistent. The pure-information magnitude is therefore fixed for the entire program.

5 Certified Status

1. Sign of the Hehl–Datta coefficient: **[P]** (relative-entropy positivity).
2. Magnitude by rigidity (given program normalizations): **[P]**.
3. Magnitude from pure information (Fisher–canonical matching): **[P]** (fixed by the null-plane limit together with absorbability on finite balls).

No residual open computation remains in the axial channel of the torsionful first law or in the information-theoretic determination of the contact coefficient. The Einstein–Cartan certification of the New Standard Model from entanglement bookkeeping is complete for finite balls in every channel.

The high-precision numerical value of the residue $\mathcal{I}_B$ itself controls only the precise size of the dictionary improvement; it is not required for the existence of the first-law equivalence or for the value of the Hehl–Datta coefficient.

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